CSL 100/ MAL 100

Tutorial 11

Structure and Union

Write a code that takes user input to insert elements into a singly linked list and then displays the list:

Enter the number of elements you want to insert into the linked list: 4

Enter element 1: 10

Enter element 2: 20

Enter element 3: 30

Enter element 4: 40

Linked List: 10 -> 20 -> 30 -> 40 -> NULL

```
#include <stdio.h>
#include <stdlib.h>
struct Node {
   int data;
   struct Node* next;
};
void insert(struct Node** head, int value) {
   struct Node* newNode = (struct
Node*)malloc(sizeof(struct Node));
   newNode->data = value;
   newNode->next = NULL;
   if (*head == NULL) {
      *head = newNode;
   } else {
      struct Node* temp = *head;
      while (temp->next != NULL) {
         temp = temp->next;
      }
      temp->next = newNode;
   }}
void display(struct Node* head) {
 if (head == NULL) {
      printf("List is empty!\n");
      return;
   }   struct Node* temp = head;
   while (temp != NULL) {
      printf("%d -> ", temp->data);
      temp = temp->next;
   }
   printf("NULL\n");
}
int main() {
   struct Node* head = NULL;
   int num, value;
   printf("Enter the number element to insert");
   scanf("%d", &num);
   for (int i = 0; i < num; ++i) {
      printf("Enter element %d: ", i + 1);
      scanf("%d", &value);
      insert(&head, value);
   }       printf("Linked List: ");
   display(head);
   return 0;}
```

Write a code to delete an element from the start (head) of a singly linked list.

Enter the number of elements you want to insert into the linked list: 4
Enter element 1: 10
Enter element 2: 20
Enter element 3: 30
Enter element 4: 40

Linked List before deletion: 10 -> 20 -> 30 -> 40 -> NULL
Node deleted from the start.
Linked List after deletion from start: 20 -> 30 -> 40 -> NULL

Take a insert and display function from previous code

```
void deleteFromStart(struct Node** head) {
    if (*head == NULL) {
        printf("List is empty, nothing to delete.\n");
        return;
    }
    struct Node* temp = *head;
    *head = (*head)->next;
    free(temp);
    printf("Node deleted from the start.\n");
}
int main() {
    struct Node* head = NULL;
    int num, value;
    printf("Enter the number of elements to insert");
    scanf("%d", &num);
    for (int i = 0; i < num; ++i) {
        printf("Enter element %d: ", i + 1);
        scanf("%d", &value);
        insert(&head, value);
    }
    printf("Linked List before deletion: ");
    display(head);

    // Delete from the start
    deleteFromStart(&head);

    // Display the linked list after deletion
    printf("Linked List after deletion from start: ");
    display(head);

    return 0;
}
```

QUIZ

Write a C program to implement a singly linked list with the following operations:

1. Insert an element at the **start** of the list (i.e., the new element should be inserted at the beginning of the list).
2. Delete an element from the **end** of the list (i.e., the last element should be removed from the list).
3. The program should allow the user to perform these operations repeatedly until they choose to exit.

Menu:
1. Insert element at the start
2. Delete element from the end
3. Display the linked list
4. Exit
Enter your choice: 1
Enter the value to insert at the start: 10

Menu:
1. Insert element at the start
2. Delete element from the end
3. Display the linked list
4. Exit
Enter your choice: 1
Enter the value to insert at the start: 20

Menu:
1. Insert element at the start
2. Delete element from the end
3. Display the linked list
4. Exit
Enter your choice: 3
Linked List: 20 -> 10 -> NULL

Menu:
1. Insert element at the start
2. Delete element from the end
3. Display the linked list
4. Exit
Enter your choice: 2
Last node deleted.

Menu:
1. Insert element at the start
2. Delete element from the end
3. Display the linked list
4. Exit
Enter your choice: 3
Linked List: 20 -> NULL

Menu:
1. Insert element at the start
2. Delete element from the end
3. Display the linked list
4. Exit
Enter your choice: 4
Exiting the program.